Report

SQL Injection (Structured Query Language)

Suyash A Gunjal

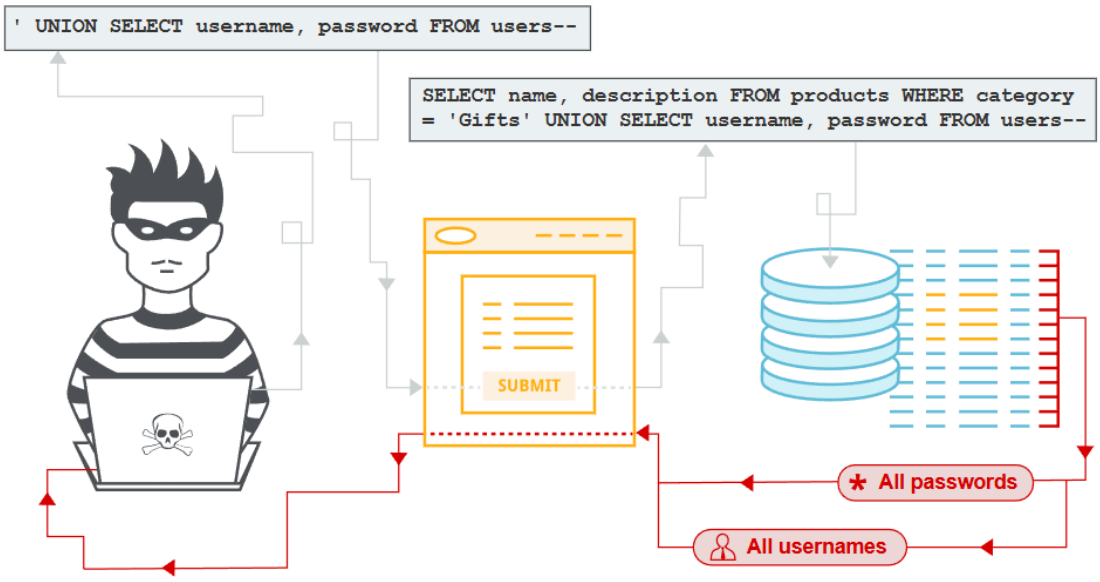
What is SQL Injection Attack?

Attackers can use SQL injection on an application if it has dynamic database queries that use string concatenation and user supplied input.

1. SQL Injection vulnerabilities are very common

2.SQL injection is a code injection technique that might destroy your database.3

3.The application's database is a frequent target for attackers because it typically contains interesting/critical data.



What is the impact of a successful SQL injection attack?

A successful SQL injection attack can result in unauthorized access to sensitive data, such as:

- Passwords.
- Credit card details.
- Personal user information.

SQL injection attacks have been used in many high-profile data breaches over the years. These have caused reputational damage and regulatory fines. In some cases, an attacker can obtain a persistent backdoor into an organization's

systems, leading to a long-term compromise that can go unnoticed for an extended period.

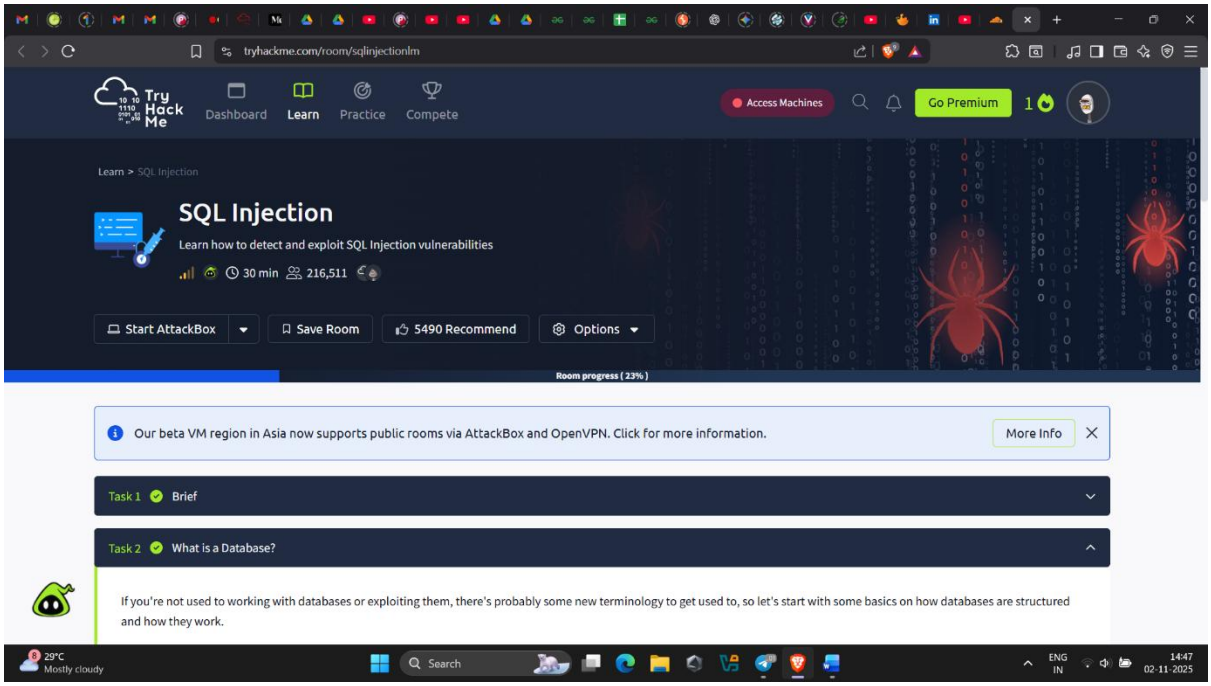
To avoid SQL injection flaws, developers need to:

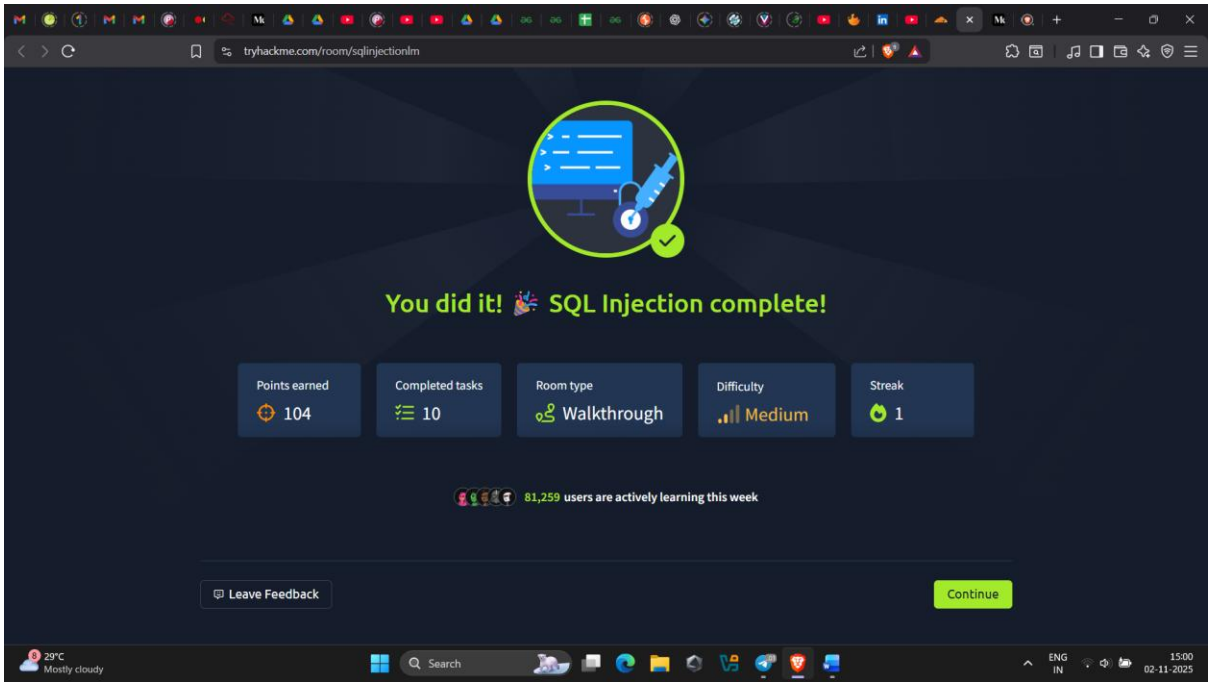
1.Stop writing dynamic queries with string concatenation or

2.Prevent malicious SQL input from being included in executed queries.

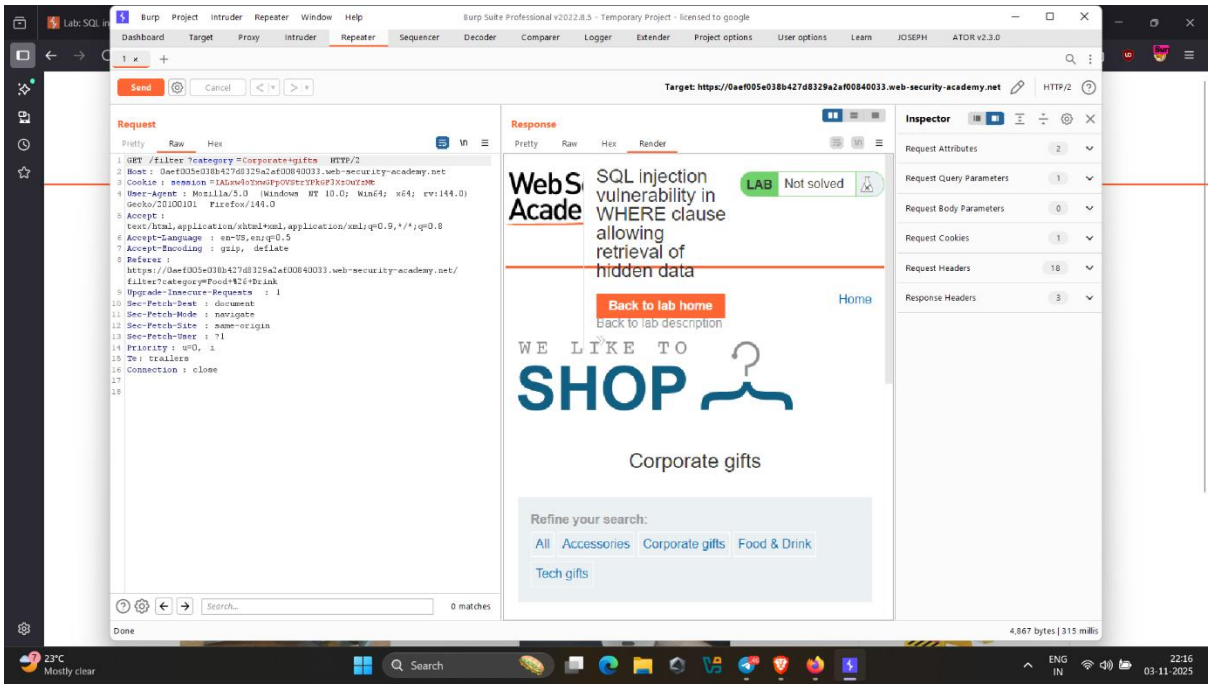
There are simple techniques for preventing SQL injection vulnerabilities and they can be used with practically any kind of programming language and any type of database. While XML databases can have similar problems (e.g., XPath and XQuery injection), these techniques can be used to protect them as well.

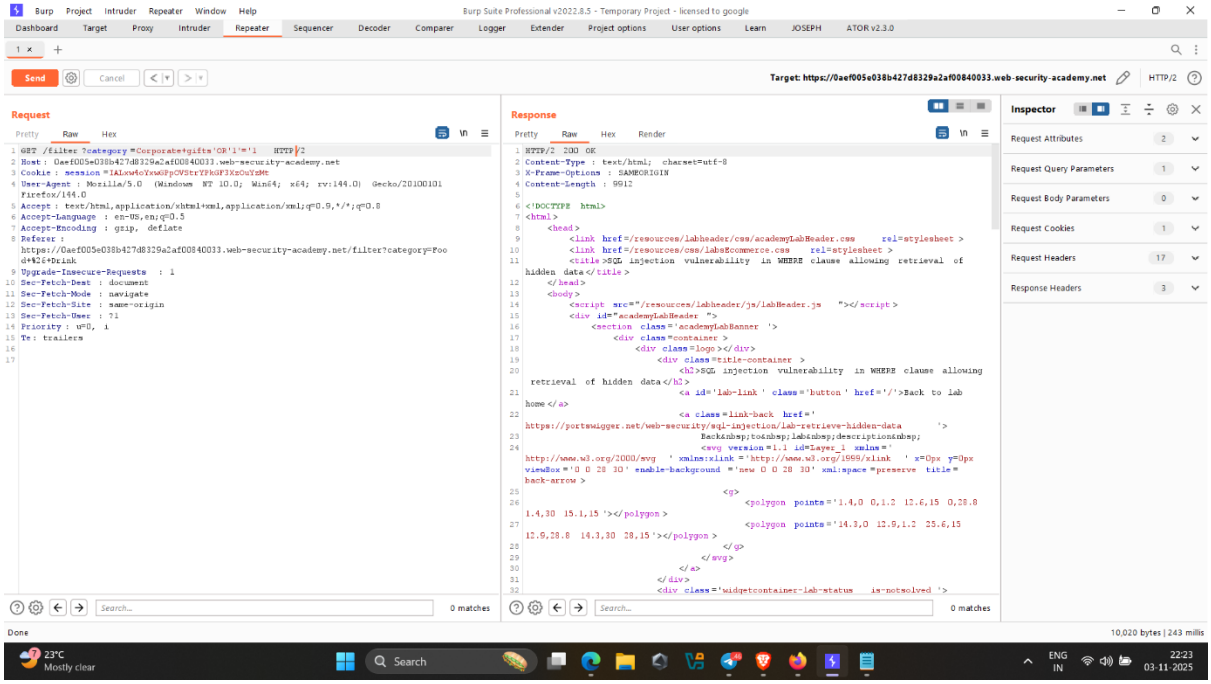
TryHackMe Lab:

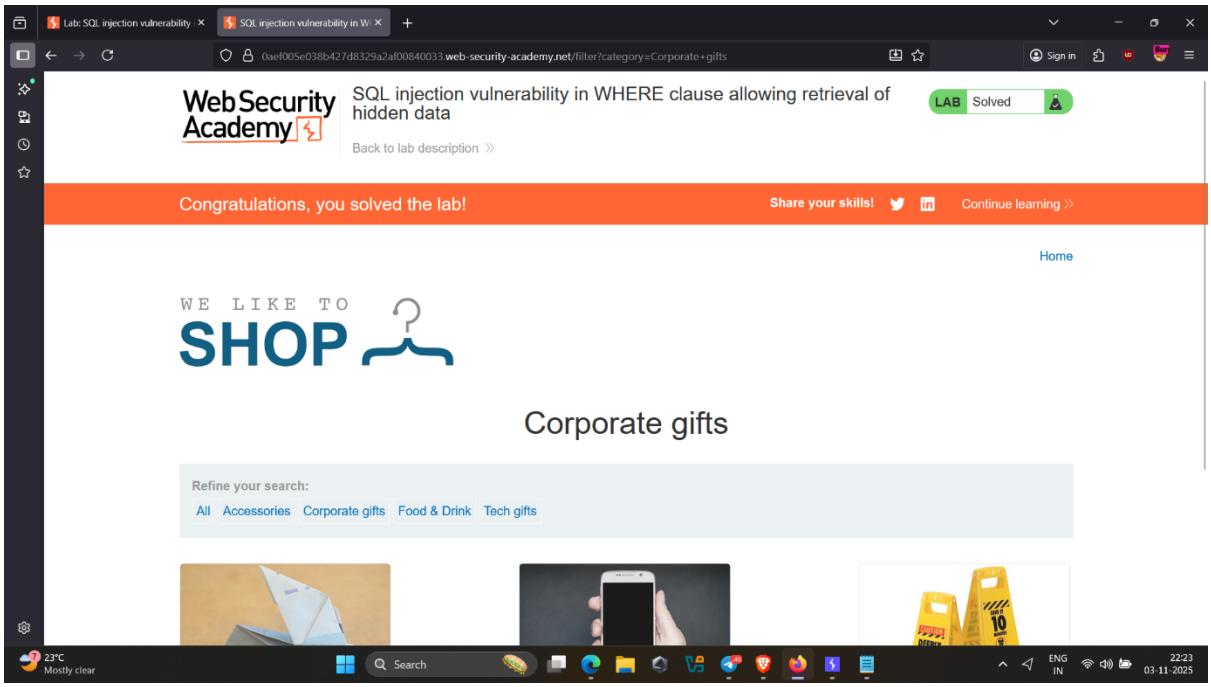




PortSwiggger Lab:







References:

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